Lesson 11: Order of Operations — Exercises

CEMC Open Courseware, Grades 7 & 8 Mathematics

1. Evaluate the following.

   (a) $3 \times 7 - 6 \div \frac{1}{3}$

   (b) $\frac{1}{3} \times (2 + 4) + 7$

   (c) $16 \times \left(\frac{3}{4} + \frac{1}{8}\right) \div 0.4$

   (d) $39 \div (0.9 + 0.4) - 2 \times (4 + 1)$

2. Evaluate using the distributive property.

   (a) $5 \times (8 - 3)$

   (b) $(6 + 4) \div 2$

   (c) $4 \times (20 + 7)$

3. Using the distributive property, Brian's solution to a problem is given below.

$$\begin{aligned} 998 \div 0.2 &= 998 \div \frac{2}{10} \\ &= (1000 - 2) \div \frac{2}{10} \\ &= 1000 \times \frac{2}{10} - 2 \times \frac{2}{10} \\ &= 200 - \frac{4}{10} \\ &= 200 - 0.4 \\ &= 199.6 \end{aligned}$$

   Is Brian's solution correct? Show how you know.

4. Remove all unnecessary brackets.

   (a) $(3 \times 5) + \left(\frac{1}{2} + \frac{5}{2}\right) \times 2 = 21$

   (b) $\Big((3 + 3 \times 3) \div 4\Big) - (27 - 24) = 0$

5. (a) Add brackets in different ways to get as many different answers as you can.

   i. $15 - 7 + 3 - 1$

   ii. $7 + 8 \times 3 - 4$

   (b) How many different answers did you get for each expression?

6. Turn the written instructions into mathematical expressions; then evaluate.

   (a) Subtract $\frac{3}{4}$ from $\frac{7}{8}$.
   Then multiply the result by 3.
   Then add 4 to this result.

   (b) Divide 7 by 0.56.
   Then add 5 to the result.
   Then multiply this result by 16.

   (c) Divide 7 by 0.25 and then add (-16).
   Subtract (-5) from (-3).
   Multiply the two results.

7. Use each of the digits 1, 2, 3, 4 (in any order) and each of the operations $+$, $-$, $\times$ (in any order) exactly once to form an expression that meets each condition. You can use brackets if needed.

   (a) An expression that equals -3.

   (b) An expression to make the largest number you can obtain.

   (c) An expression to make the smallest number you can obtain.